# Security risk assessment report

| Part 1: Select up to three hardening tools and methods to implement |
| --- |
| 1.**Implement Strict Password Policies:** Enforce policies that prohibit password sharing, restrict the use of default credentials, and require secure hashing algorithms for stored passwords.<br>**2. Configure Firewall Rules (Access Control Lists):** Implement explicit inbound and outbound traffic filtering rules on all corporate firewalls.<br>**3. Enforce Multi-Factor Authentication (MFA):** Require all employees and database administrators to verify their identity using a second factor (e.g., OTP via mobile app or physical security key) before accessing the network or internal databases. |

| Part 2: Explain your recommendations |
| --- |
| **1. Password Policies:** The scenario highlights that employees are sharing passwords and the database administrator is using a default password. Establishing and enforcing a strict password policy is an effective hardening technique because it immediately addresses these critical human-error vulnerabilities. It ensures that default credentials are automatically rejected and individual accountability is maintained. This policy should be enforced continuously through IAM systems.<br>**2. Firewall Rules:** The organization's firewalls currently lack filtering rules. Configuring strict Access Control Lists (ACLs) is highly effective because it directly prevents unauthorized external traffic from entering the network and stops unauthorized internal data from being exfiltrated. Firewall rules act as the first line of network defense and should be continuously monitored, with rules reviewed and updated on a monthly or quarterly basis.<br>**3. Multi-Factor Authentication (MFA):** Relying solely on passwords has already led to a major data breach for this organization. MFA is an essential hardening method because even if an attacker successfully guesses a default password or intercepts a shared password, they still cannot access the system without the secondary physical or biometric token. MFA should be configured once during user onboarding and maintained continuously for every login attempt. |